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EU AI Act

Slovakia

AI Ecosystem Brief

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Acknowledgements

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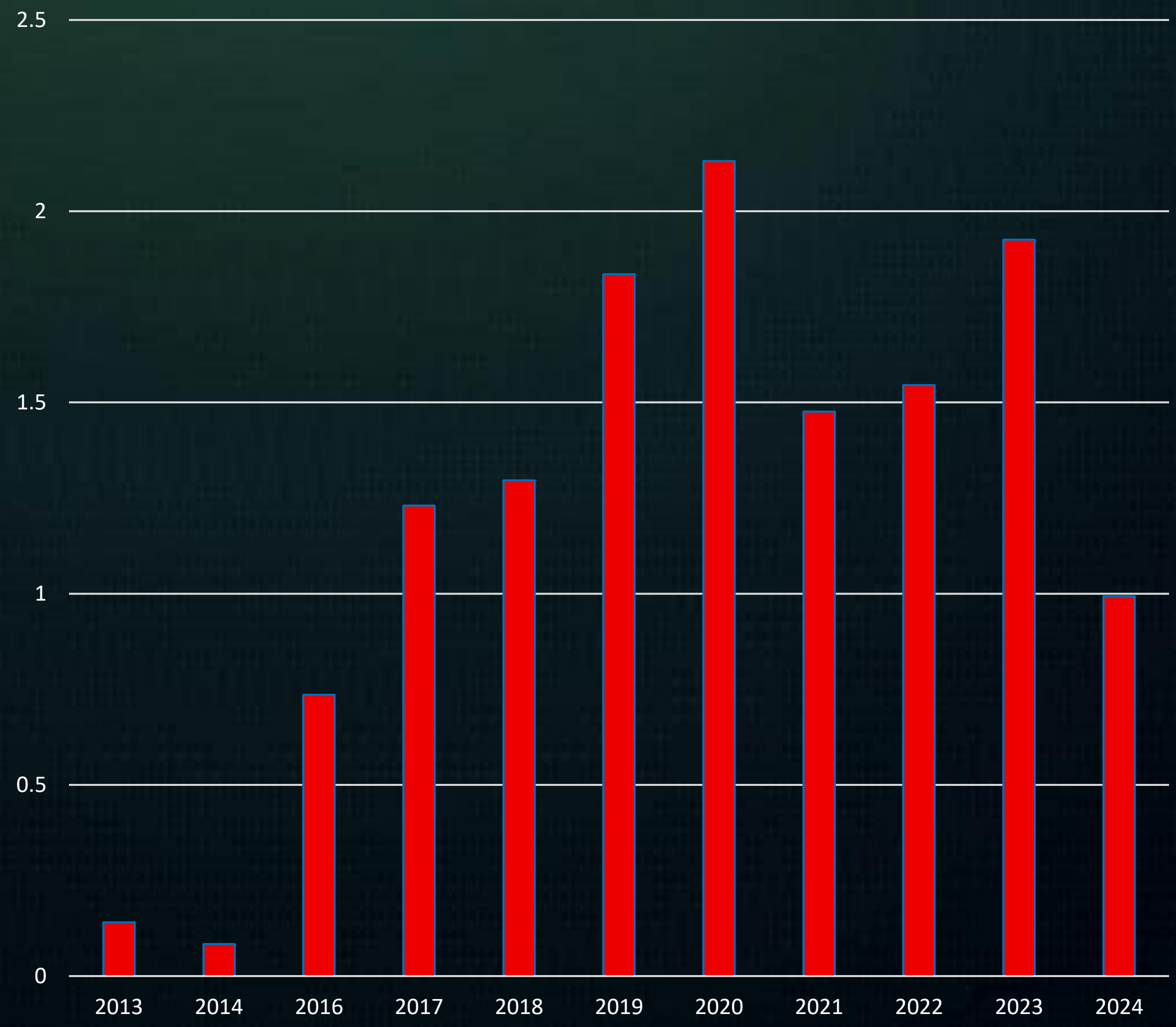
Sustaining Slovakia's AI Talent Momentum: Strategic Priorities to Convert Irregular Growth into a Stable, Competitive, and Innovation-Driven Ecosystem by 2030

Evolution of AI questions and answers in Slovakia (November 2025)



Explanation: This chart shows the number of questions and answers on Stack Overflow in Slovakia.

The evolution of AI-related activity in Slovakia, measured through the frequency of AI questions and answers between 2013 and 2024, shows a pattern of **early stagnation**, followed by **strong acceleration**, and then a **notable post-2020 fluctuation**. Values remain very low and nearly flat through 2013–2014 (0.07 and 0.04), indicating a period in which Slovakia's AI community, research presence, and public engagement were still nascent.

A first major inflection occurs in **2016**, where activity increases sharply to **0.368**, followed by sustained growth through **2017–2020**, culminating in a peak of **1.066 in 2020**. This period corresponds to Slovakia's strengthening digital-governance agenda, the expansion of university programmes in informatics and automation, and the emergence of multi-stakeholder platforms such as AISlovakIA. The availability of EU structural funds and the initial wave of digital-transformation priorities further amplified interest in AI-related topics.

After 2020, however, the data shows **non-linear behaviour**: activity drops to **0.738 in 2021**, recovers to **0.773 in 2022**, rises again to **0.963 in 2023**, and then falls significantly to **0.496 in 2024**. This volatility suggests that Slovakia's AI talent and skills ecosystem is still in an **early developmental stage**, sensitive to policy cycles, funding availability, labour-market dynamics, and the uneven uptake of AI across industries.

The oscillation contrasts with countries with deeper AI infrastructures and indicates structural constraints typical of emerging ecosystems:

- A narrow advanced-skills base concentrated in a few universities and research centres.
- Limited domestic R&D intensity and dependence on EU-funded projects.
- Insufficient demand-side pull from industry, especially among SMEs.
- Early-stage institutional coordination for AI governance and innovation.

Still, the fact that Slovakia achieved repeated peaks and did not revert to pre-2016 lows demonstrates the **maturing of its AI community**, growing societal interest, and gradual strengthening of digital-skills pipelines. To convert this uneven progress into sustainable competitiveness, Slovakia should pursue the following priorities:

1. **Expand advanced talent pipelines**
    - Increase specialised AI master's and PhD fellowships.

- Embed AI and data-science content into STEM and engineering programmes.
- Strengthen dual-education models and industry-linked practical training.

2. **Deepen research–industry integration**
   - Incentivise Slovak firms to adopt AI solutions developed by domestic universities and research centres.
   - Support links between EDIHs, innovation hubs, and industrial clusters.
   - Prioritise applied research in mobility, manufacturing, public administration, and healthcare.
3. **Create long-term national AI R&D programmes**
   - Complement EU-funded initiatives with dedicated national funds for AI experimentation, trusted-AI development, and HPC-enabled research.
   - Support development of testbeds and data-sharing environments.
4. **Attract and retain international AI experts**
   - Promote Slovakia's affordability and quality of life.
   - Offer competitive research contracts and mobility schemes.
   - Use the AI sandbox and digital-innovation hubs as magnets for global talent.
5. **Establish a national AI talent monitoring framework**
   - Create an annual Slovakia AI Talent Index tracking skills supply, workforce movements, industry demand, regional disparities, and research output.
   - Use data to anticipate bottlenecks and guide education and labour-market policy.

In summary, Slovakia's AI activity trajectory reveals meaningful progress but also significant volatility. The country has moved beyond its early-stage baseline, showing capacity to generate spikes in AI engagement and interest. Nonetheless, the inconsistent pattern highlights structural vulnerabilities: reliance on external funding cycles, limited absorptive capacity in industry, and an AI talent pool that has not yet reached critical mass. Building a resilient AI ecosystem by 2030 will require Slovakia to stabilise talent growth, expand advanced skills pathways, strengthen the research–industry interface, and secure long-term investment. These steps are essential for translating intermittent progress into sustained, innovation-driven competitiveness within the broader European AI landscape.

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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems are unknown

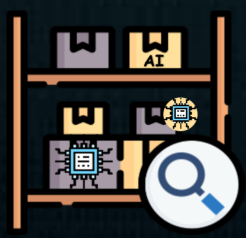
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.

Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.

Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contents

This section outlines the structure of the report, detailing chapters on Slovakia’s AI policy landscape, regulatory initiatives, governance mechanisms, findings, and recommendations. It serves as a navigation guide, enabling readers to quickly locate specific analyses, datasets, and policy discussions relevant to Slovakia’s evolving artificial intelligence ecosystem.

Overview

Artificial Intelligence (AI) is increasingly shaping Slovakia's digital future, and the country is steadily aligning itself with European Union standards for ethical, rights-based, and innovation-driven AI governance.

Anchored in the EU AI Act and Slovakia's 2030 Digital Transformation Strategy, Slovakia is adopting a supervisory-first, values-centered model that prioritizes regulatory clarity, institutional oversight, and human-centric innovation.

Rather than introducing sweeping standalone AI legislation, Slovakia is embedding AI governance into its broader digital modernization agenda. A proposed Slovak AI Act is expected to define the competences of domestic institutions under the EU AI Act—particularly regarding oversight, enforcement, and risk categorization. This foundational step reflects Slovakia's intention to create a structured, EU-compliant regulatory environment for trustworthy AI deployment across sectors.

At the core of Slovakia's AI regulatory architecture are the Office for Personal Data Protection (ÚOOÚ) and the Public Defender of Rights (VOP)—both designated as competent authorities to monitor AI's impact on fundamental rights. Their roles are expected to expand as Slovakia matures its compliance framework and establishes coordinated supervision across government institutions.

While Slovakia has not yet launched a dedicated AI agency akin to Spain's AESIA or implemented a regulatory sandbox, the government envisions Innovation Laboratories within public administration to experiment with governance tools, policy mechanisms, and AI technologies in real-world environments. This demonstrates a cautious but forward-looking approach to fostering responsible experimentation. Slovakia's AI governance is supported by a growing body of sectoral ethics instruments, such as the 2022 Ethical Guidelines for Teachers, and internal directives by institutions like Comenius University.

These soft-law initiatives reflect an early commitment to embedding transparency and human oversight in AI use—especially in education and legal practice. From a strategic standpoint, AI is identified as a key enabler of economic competitiveness in Slovakia's 2030 Digital Transformation Strategy. The country seeks to reduce reliance on low-value-added production by leveraging AI in manufacturing, smart public services, and data-driven administration. High-priority areas include the modernization of public governance, upskilling of the workforce, and boosting digital infrastructure.

Slovakia's AI readiness is further demonstrated by its exceptionally high score in the Data & Infrastructure pillar of the Government AI Readiness Index (80.91), well above both regional and global averages. Strong broadband coverage, expanding 5G infrastructure, and a growing emphasis on high-performance computing and open data platforms provide a robust foundation for AI adoption.

Nevertheless, the Technology sector score (41.40) highlights ongoing challenges in innovation scale-up, private sector R&D, and talent retention. Slovakia continues to face constraints in deploying advanced AI systems at scale, but policy interventions—such as the Digital Coalition, targeted STEM education, and support for SMEs—seek to close this gap over time. Slovakia's participation in international dialogues—such as the EU Coordinated Plan on AI, the Digital Europe Programme, and the OECD AI Principles—demonstrates its commitment to globally harmonized standards and interoperable digital governance. The country is actively modelling its strategy on digital leaders like Finland, France, and the UK.

In summary, Slovakia is developing a measured, EU-integrated AI governance model that emphasizes fundamental rights, digital inclusion, and institutional accountability.

Specific AI Governance or Law

Slovakia has not introduced standalone AI legislation but is implementing the EU AI Act. A national AI Act has been proposed to define the roles of Slovak authorities under the EU framework. As of the EU AI Act's 2 November 2024 deadline, Slovakia designated the Office for Personal Data Protection (ÚOOÚ) and the Office of the Public Defender of Rights (VOP) as key supervisory bodies. Further designations may follow to expand oversight capacity.

Specific AI Governance in Slovakia

Slovakia's emerging AI oversight model is rooted in compliance with the EU AI Act, with initial supervisory responsibilities assigned to ÚOOÚ and VOP. As the AI regulatory landscape evolves, it is expected that these bodies will be joined by other public institutions, establishing a coordinated supervisory framework.

Looking forward, Slovakia's proposed national AI Act and its engagement with EU-level policy developments will shape the country's trajectory in AI governance. While currently in a preparatory phase, Slovakia's approach suggests a gradual but deliberate strategy to ensure AI systems are used safely, ethically, and in a way that protects fundamental rights across sectors.

AI Ethics Guidelines and Sectoral Governance Tools

While Slovakia does not yet have a comprehensive national AI strategy, the country has introduced sector-specific ethical guidance, notably the Ethical Guidelines for Teachers issued in 2022. These guidelines offer recommendations on responsible AI use in educational settings, reflecting growing public sector interest in AI governance principles such as transparency and human oversight.

Additional sectoral initiatives include the Comenius University in Bratislava's internal directive on AI usage in education, as well as a proposal from the Slovak Bar Association to establish internal rules for legal associates using AI tools. These initiatives mark early efforts to address the ethical and professional dimensions of AI within specialized domains. Currently, AI is not specifically addressed under Slovakia's intellectual property or data protection laws, and no soft law instruments or guidelines exist in these areas beyond general EU-level applicability. Nonetheless, regulatory authorities and public institutions are preparing to adapt existing frameworks to the unique challenges AI presents.

To date, there is limited precedent in AI enforcement in Slovakia. However, in a notable judicial decision, the Regional Court in Trenčín (21 November 2024, Case No. 19CoPr/2/2024) emphasized the role of human judicial reasoning. The court recognized the utility of AI tools in improving efficiency and access to legal information but affirmed that final legal judgments must rest with human judges, not automated systems. This underscores a cautious and rights-based approach to AI integration in sensitive sectors like justic

Slovakia’s National AI Strategy

Slovakia’s AI governance is shaped by a value-driven, EU-integrated approach that places emphasis on digital transformation, ethical public administration, and economic modernization. Anchored in its 2030 Digital Transformation Strategy and supported by ongoing legislative efforts, Slovakia is aligning itself closely with the EU AI Act while setting the foundations for national oversight and sectoral adaptation of AI technologies.

Slovakia’s AI governance trajectory is rooted in principles consistent with EU and OECD values, including human-centricity, digital equity, and innovation-led growth. While Slovakia currently lacks a standalone AI law, its approach is evolving through the convergence of strategic planning—chiefly the 2030 Digital Transformation Strategy for Slovakia—and its alignment with EU legislation such as the AI Act and Digital Europe Programme. A defining element of Slovakia’s strategy is the emphasis on societal inclusion, ethical standards, and infrastructure preparedness. AI is explicitly recognized as a key enabling technology within the national digital strategy, alongside IoT, big data, blockchain, and high-performance computing. These technologies are seen as the cornerstone of Slovakia’s transition to a knowledge-based and data-driven economy. While Slovakia has proposed a national AI Act primarily to designate responsibilities among domestic authorities under the EU AI Act (Articles 28–29 and 70), broader policy execution is supported by ethical and educational instruments. For instance, Ethical Guidelines for Teachers on AI Use (2022) and institutional directives by Comenius University provide early sectoral governance examples rooted in public trust and responsible use. Slovakia's model also places strong emphasis on regulatory coherence and institutional coordination. In line with EU obligations, Slovakia has designated the Office for Personal Data Protection (ÚOOÚ) and the Public Defender of Rights (VOP) as supervisory bodies for AI. Additional authorities are expected to be assigned to ensure comprehensive oversight.

Judicial developments reinforce Slovakia's rights-based approach. In a 2024 ruling, the Regional Court in Trenčín affirmed that while AI can aid judicial efficiency, final legal decisions must remain under the jurisdiction of human judges—highlighting the limits of automation in sensitive democratic functions. Slovakia's approach to sectoral integration remains incremental but promising. The Digital Transformation Strategy outlines strategic goals across multiple domains, including education, public services, and science. However, unlike Spain's AESIA or regulatory sandbox, Slovakia has not yet implemented dedicated AI testing environments. Still, it envisions "innovation laboratories" within public administration to pilot new technologies and regulatory concepts. The strategy also focuses on AI readiness through human capital development and infrastructure modernization. It emphasizes the need for comprehensive reform in education, digital skill-building, and requalification programs, targeting both general digital literacy and advanced technological proficiency. The establishment of a Digital Coalition and various STEM initiatives reflect efforts to enhance societal preparedness for the AI era.

From a governance perspective, Slovakia champions cooperative coordination—a model involving cross-sectoral engagement and public-private collaboration. Despite lacking a fully mature AI ecosystem, the state's structured approach, underpinned by EU co-financing mechanisms and interoperability frameworks, provides a solid base for future AI deployment.Slovakia's framework remains a dynamic and adaptive one. The 2030 strategy underscores the need for continuous evaluation of regulatory performance, responsiveness to global trends, and integration of best practices from countries like Finland, France, and the UK. The country's recognition of AI as a key growth engine signals its readiness to evolve toward a more innovation-intensive, inclusive, and secure digital society.

Approach to AI Regulation

Slovakia is taking measured steps to align with the European Union's framework for artificial intelligence governance, focusing on institutional preparedness, sector-specific initiatives, and regulatory harmonization. While it has not yet enacted standalone AI legislation, Slovakia is laying the groundwork for national oversight through targeted proposals and the implementation of key EU AI Act provisions.

1. Regulatory and Strategic Frameworks

Slovakia's AI governance approach is primarily rooted in alignment with the EU AI Act, which currently serves as the overarching legal framework applicable in the country. As of now, Slovakia has not enacted standalone AI legislation. However, a draft Slovak AI Act is under consideration. Its primary aim is to define the competences of national public authorities—particularly with respect to oversight functions outlined in Articles 28–29 and Article 70 of the EU AI Act. Although no comprehensive national AI strategy has been adopted, ethical principles are beginning to emerge in sectoral settings, reflecting a cautious but deliberate policy shift toward responsible AI usage. These developments highlight Slovakia's intent to build a foundational governance structure in tandem with EU-level regulations.

2. Institutional Support and Strategic Missions

Slovakia has officially designated two public authorities to oversee compliance with fundamental rights in relation to AI systems, in accordance with EU AI Act obligations. These are the Office for Personal Data Protection of the Slovak Republic (ÚOOÚ) and the Office of the Public Defender of Rights (VOP). Their roles are expected to expand as domestic AI supervision mechanisms mature. Further institutional development is likely in the coming months, as Slovakia anticipates broadening the scope of its supervisory bodies. These institutions will play a central role in coordinating national efforts, ensuring legal compliance, and promoting human-centric AI deployment.

3. Standards Development and Global Engagement

While Slovakia has not yet published national AI standards or joined global AI standardization forums in a formal capacity, its regulatory development remains strongly tied to EU-level initiatives, such as the Coordinated Plan on AI (2019–2027) and forthcoming implementation guidelines from the European Commission.

4. Sectoral and Extended Regulatory Approaches

Slovakia’s sector-specific engagement with AI remains in early stages. A noteworthy development includes the Ethical Guidelines for Teachers issued in 2022, addressing the responsible use of AI in educational contexts. Additionally, the Comenius University in Bratislava has adopted an internal directive on AI use in academia, while the Slovak Bar Association has proposed internal rules concerning AI usage among legal associates. Beyond these initiatives, AI is not yet explicitly addressed in existing Slovak intellectual property, data protection, consumer protection, or labor laws. However, as the EU AI Act begins to influence member states, Slovakia is expected to integrate AI-related considerations across a broader array of regulatory domains.

5. Fostering Innovation and Indigenous Development

While Slovakia does not yet operate a formal regulatory sandbox akin to Spain’s RD 817/2023, sector-specific experimentation with AI—particularly in legal, educational, and public administration contexts—is gradually increasing. These developments suggest a cautious but intentional move toward supporting safe AI innovation. The absence of formalized public-private AI development programs or R&D incentives has so far limited domestic AI capacity-building. However, with the national AI Act under development and supervisory structures in place, Slovakia is poised to expand its role in fostering responsible AI innovation within the EU framework.

6. AI in Public Sector Transformation

AI adoption in Slovakia's public sector remains limited, with existing efforts largely focused on guiding ethical use in sensitive domains like justice and education. A notable judicial decision by the Regional Court in Trenčín (21 November 2024) underscored the principle that, despite AI's potential benefits in speeding up legal processes and data comparison, final judgments must be rendered by human judges, affirming the constitutional value of human oversight. This case signals a commitment to fundamental rights and legal integrity, and offers a precedent for future debates about AI's role in the justice system and other public services.

7. AI Skills, Education and Inclusion

Formal initiatives to build AI-related skills and foster digital inclusion are currently limited in Slovakia. However, sector-specific efforts, such as educational guidelines and university-led policies, indicate growing awareness of the need for AI literacy. These initiatives, while fragmented, may serve as precursors to a national skills strategy in the future. Slovakia's approach to inclusive AI development remains largely undeveloped, but EU programs and funding instruments may serve as catalysts for further engagement in this area.

8. Infrastructure, Compute and Data Foundations

At present, Slovakia is still in the early stages of building the infrastructure required for AI at scale. Unlike countries with established regulatory sandboxes or national data governance frameworks, Slovakia has not yet created formal environments for secure AI experimentation or data sharing. Future development in this area will likely be driven by EU-funded digital infrastructure projects, including those focused on high-performance computing, interoperable data ecosystems, and trusted data spaces.

In summary, Slovakia is gradually developing an AI governance framework aligned with the EU AI Act, prioritizing regulatory preparedness, ethical guidance, and institutional designation. While national strategies and infrastructure are still emerging, Slovakia's evolving legal and policy landscape sets the stage for more structured, rights-based, and sector-aware AI governance in the years ahead.

Approach to Foster Innovation

Slovakia's strategic approach to AI and digital transformation is anchored in EU alignment, cross-sectoral modernization, and future-readiness. Through the 2030 Digital Transformation Strategy, Slovakia aims to integrate AI and emerging technologies into economic, societal, and public service innovation while safeguarding democratic values, inclusive growth, and institutional trust.

1. Unlocking Economic Value through Targeted AI Deployment

Slovakia's 2030 strategy views artificial intelligence as a core enabler of economic transformation, particularly in manufacturing, public services, education, and regional development. Recognizing AI's value across sectors, Slovakia aims to modernize its industrial base, shift towards a knowledge- and data-driven economy, and reduce its dependence on low-value-added production. AI is intended to boost productivity, transparency, and competitiveness, especially in high-impact areas such as public administration, education systems, and smart regional development.

2. Catalysing Innovation through Strategic Clustering and Ecosystems

Although Slovakia currently lacks a dense network of operational Digital Innovation Hubs (DIHs), its strategy outlines plans to build innovation ecosystems through public-private cooperation, smart industry support, and regional smart city initiatives. The capital, Bratislava, is positioned as a regional innovation hub within the V4 countries. The strategy promotes the setup of Innovation Laboratories to test new digital policies, governance models, and AI-enabled technologies within public institutions, laying the groundwork for future clustering.

3. Enabling Trusted and Responsible Data Use

Slovakia identifies data access and governance as essential to digital advancement. The strategy highlights the need for interoperable infrastructure, open government data, and regulatory alignment with EU standards such as the GDPR and EU AI Act. Although legislative adaptation is still evolving, Slovakia promotes data security, privacy, and trust as foundational to future AI deployments and citizen engagement in digital systems.

4. Scaling AI in the Public Sector

Public administration is a strategic priority in Slovakia's digital transformation. The strategy calls for digitally skilled governance, reduced administrative burdens, and AI-based decision support systems. Early sectoral examples include AI integration in education (teacher guidelines, university directives) and judicial oversight—as reflected in a 2024 court ruling reaffirming the need for human interpretation in legal decisions, even when AI is used for data analysis.

5. Advancing Compute and Research Infrastructure

Slovakia underscores the importance of infrastructure including high-performance computing (HPC), 5G deployment, and fiber-optic networks as prerequisites for scalable AI. While investment in compute and research infrastructure lags behind EU frontrunners, the strategy includes commitments to co-financing digital R&D, attracting international partnerships, and aligning with EU initiatives such as Digital Europe Programme and Connecting Europe Facility.

6. Empowering SMEs and Startups

Recognizing its strong industrial base and entrepreneurial potential, Slovakia targets support for SMEs through smart industry action plans, regulatory simplification, and sector-specific digital programs. The strategy calls for venture capital attraction, support for start-ups, and SME requalification, aiming to make innovation more accessible across business sizes and sectors. AI uptake by SMEs is considered vital for closing Slovakia's productivity gap.

7. Developing a Future-Ready Workforce

Human capital is central to Slovakia’s digital vision. The strategy emphasizes educational reform, lifelong learning, and targeted investment in STEM and digital skills. Initiatives like the Digital Coalition and the Education Informatization Programme aim to close the digital skills gap and modernize curricula. However, Slovakia continues to face talent drain, regional disparities, and a shortage of advanced digital professionals, particularly in AI-related fields.

8. Aligning AI Innovation with Sustainability Goals

Slovakia integrates sustainability into its digital transformation through a dual focus on environmental efficiency and green AI adoption. The strategy aligns with the 2030 Agenda for Sustainable Development and European Green Deal goals, advocating for the use of digital technologies—including AI—in climate adaptation, mobility, smart infrastructure, and circular economy initiatives.

9. Building Public Trust and Ethical AI

Ethical oversight and democratic values form the backbone of Slovakia’s digital transition. The strategy promotes transparency, citizen-centricity, and rights-based AI use. Early guidelines—such as ethical recommendations for AI in education—set precedent for future sectoral ethics codes. Institutional trust is reinforced through judicial rulings and the assignment of AI supervisory roles to authorities like the Office for Personal Data Protection (ÚOOÚ) and the Public Defender of Rights (VOP).

10. Shaping Global AI Governance and Standards

Slovakia’s alignment with EU regulatory frameworks ensures it contributes to shaping global AI norms via coordinated EU policymaking. The strategy is inspired by digital leadership in countries like Finland, France, Singapore, and the United Kingdom, and encourages Slovakia’s participation in OECD, UN, and EU-level digital governance dialogues. As the EU AI Act enters into force, Slovakia is poised to support the standardization of human-centric, trustworthy AI.

Digital Decade 2025: Report

Slovakia remains committed to achieving the 75% business AI adoption rate by 2030, aligning with the EU-wide objective. Progress so far has been promising and appears consistent with the country’s national trajectory. However, as adoption grows, the rapid increase observed between 2023 and 2024 may begin to slow.

Slovakia's strategy places strong emphasis on promoting AI use among enterprises, supported by a variety of planned initiatives. These are complemented by education and skills development measures in the AI domain. Key actions in 2024 included setting up an AI information hub, launching programmes to improve AI literacy among non-ICT professionals, preparing a public AI awareness campaign, and continuing active involvement in EU-level AI initiatives.

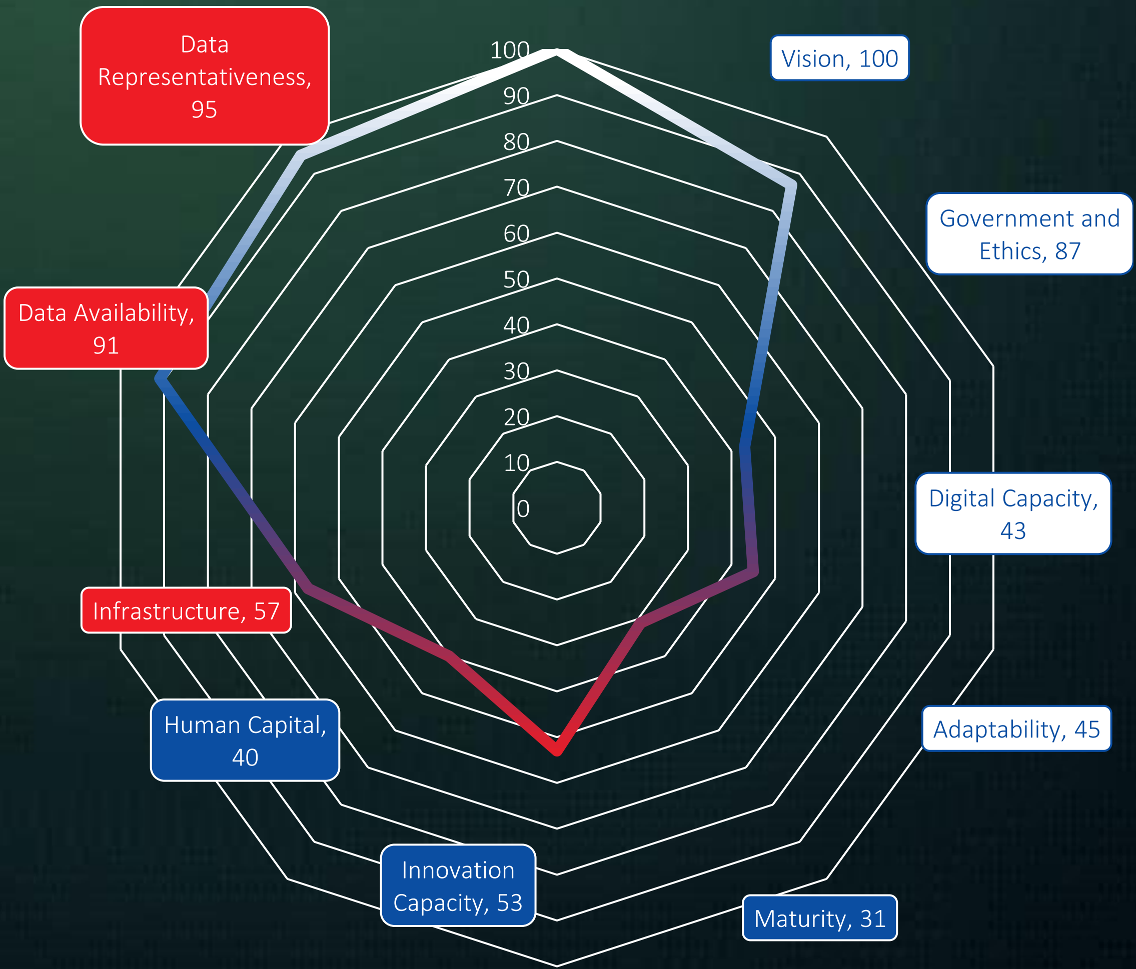
Despite these efforts, the 2030 objective is highly ambitious and will demand sustained commitment and enhanced efforts. While the updated roadmap did not include new actions, authorities indicated openness to future measures aimed at supporting AI adoption. Slovak institutions have also highlighted persistent informational and managerial obstacles, such as limited awareness of AI benefits and resistance to change due to outdated practices and weak digital capabilities. Low overall levels of business digitalisation also present a barrier to adopting advanced technologies like AI.

The 2024 recommendations on AI, cloud, and data analytics advised Slovakia to facilitate the development, deployment, and uptake of advanced technologies, including through capital investments. In response, Slovakia implemented some policy measures in 2024. While these focused more broadly on enhancing business digitalisation rather than directly supporting AI or data analytics, they nonetheless contribute to the recommendation's goals. Initiatives like the Digital Policy Forum and ALT-EDIC participation are designed to stimulate innovation by encouraging collaboration between the public and private sectors. Additionally, the continuation of European Digital Innovation Hubs (EDIHs) and the Digitrans project aim to help especially SMEs adopt and utilise digital tools effectively.

Government AI Readiness

Eastern Europe ranks 4th in this year's Government AI Readiness Index, with an average score of 57.88—over 10 points above the global average (47.59). Estonia (72.62) leads the region and is the only Eastern European country to feature in the global top 25. The region performs above the global average across all three pillars. Its strongest performance is in the Data & Infrastructure pillar (72.10), making it the third-best performing region globally in this area. Eastern Europe also shows solid results in the Government pillar (62.52), where it again ranks third worldwide. However, the Technology Sector remains a challenge, with a regional score of 39.03—more than 20 points behind the other two pillars—and ranking fifth globally, falling narrowly behind the Middle East and North Africa (39.34). This reflects the need for further investment in technological capacity and innovation to unlock the region's full AI readiness potential.

2024 Scores by Dimension



Government

Slovakia scores 68.76 in the Government pillar, reflecting its coordinated and EU-aligned approach to AI governance. The country does not yet have a standalone AI law but is actively working to transpose and implement the EU AI Act through a proposed Slovak AI Act. This forthcoming legislation focuses on defining the roles and responsibilities of national authorities in compliance with provisions of the EU AI Act, particularly Articles 28–29 and Article 70. Slovakia has designated the Office for Personal Data Protection (ÚOOÚ) and the Public Defender of Rights (VOP) as competent supervisory authorities, laying the groundwork for a robust national oversight framework in line with European regulatory standards.

Technology Sector

Slovakia’s score of 41.40 in the Technology pillar signals an evolving innovation environment with clear potential. The 2030 Digital Transformation Strategy recognizes AI as a critical enabler of economic and societal modernization. However, like much of the Eastern European region, Slovakia faces challenges in scaling advanced AI research and industrial capabilities. Current focus areas include building digital skills, incentivizing entrepreneurship, and fostering public-private R&D collaborations. While AI solution creation at scale remains limited, Slovakia shows promise in sectors such as smart manufacturing, legal technology, and educational technology, particularly through pilot initiatives and academic engagement.

Data & infrastructure Sector

With a score of 80.91, Slovakia demonstrates exceptional strength in the Data & Infrastructure pillar—well above both the regional and global averages. The national strategy emphasizes the development of high-performance computing, interoperability frameworks, and open data platforms. The government's commitment to broadband expansion, fiber-optic infrastructure, and 5G network deployment underpins its broader vision for a connected and innovation-ready society. The presence of strong data governance practices and emerging efforts to establish digital innovation hubs (DIHs) reinforces Slovakia’s infrastructure maturity.

With a total score of 63.69, Slovakia is one of the stronger performers in Eastern Europe on the Government AI Readiness Index. The country benefits from a solid regulatory framework, high institutional preparedness, and world-class data infrastructure. While technological innovation and private-sector AI capacity are still in development, Slovakia's alignment with EU digital policies and its comprehensive 2030 strategy provide a clear roadmap for continued growth. The next phase will depend on the effective implementation of national legislation under the EU AI Act, targeted investment in AI capabilities, and continued focus on building a future-ready workforce. Together, these efforts position Slovakia as a rising actor in the European AI ecosystem

Conclusion

Slovakia's artificial intelligence ecosystem is steadily evolving—anchored in EU-aligned governance and a strategic commitment to digital modernization. With institutional preparedness rooted in the forthcoming Slovak AI Act and supervisory designations under the EU AI Act, Slovakia is building a framework for ethical, rights-based AI deployment. The country's alignment with EU digital strategies and its 2030 Digital Transformation Strategy reflects a deliberate move toward responsible and inclusive innovation.

Slovakia ranks highly on data availability and digital infrastructure, bolstered by EU-backed investments and national priorities like broadband expansion, 5G deployment, and interoperable government data platforms. This infrastructure strength positions Slovakia to scale AI adoption in education, justice, and public administration—areas already testing early use cases through ethical guidelines and judicial rulings. The governance model is grounded in constitutional values, with authorities such as the Office for Personal Data Protection (ÚOOÚ) and the Public Defender of Rights (VOP) overseeing compliance with fundamental rights.

Yet significant challenges remain. Slovakia's comparatively modest technology sector score highlights ongoing barriers to AI commercialization, industrial integration, and research scalability. The absence of a regulatory sandbox and limited public-private R&D pipelines have constrained rapid innovation cycles. Addressing this requires greater support for SME participation, expansion of compute capacity, and enhanced university–industry collaboration to translate Slovakia’s strong academic base into entrepreneurial momentum.

To realize its AI potential, Slovakia must deepen regional ecosystem development beyond Bratislava and invest in high-impact talent pipelines. Initiatives like the Digital Coalition and Education Informatization Programme are steps in the right direction but require broader reach and sustained funding to combat talent shortages and regional disparities. Further, while Slovakia’s legal and ethical frameworks are principled, they must continuously adapt to the demands of frontier technologies.

Internationally, Slovakia’s harmonization with EU digital governance and active engagement in regional dialogues give it a foothold in shaping cross-border AI standards. As the EU AI Act enters implementation, Slovakia has an opportunity to contribute a model of value-driven, coordinated governance that reflects democratic integrity and social resilience.

If met with focused investment, legal coherence, and institutional trust, Slovakia can emerge not just as a regulatory adherent—but as a credible contributor to Europe’s AI future.

Appendix A: Expert Vignettes

This appendix presents short expert vignettes that illustrate how Slovakia's institutional landscape for AI oversight, coordination and policy implementation operates in practice. Each vignette is written from the perspective of a key AI ecosystem typical actor) and highlights institutional responsibilities, accountability arrangements and multi-stakeholder mechanisms that shape trustworthy, inclusive and adaptive AI governance

Juraj Sasko, *Partner*, Visibility

The 2030 target for widespread AI adoption in Slovak companies is strategically correct but extremely ambitious given the current low level of digital maturity across much of the SME segment. Based on long-term experience with digital transformation in online marketing, reaching this goal will depend less on regulation and more on practical implementation support, access to talent, and clear, measurable business ROI from AI deployments.

Biography

Juraj Sasko, began his journey in online marketing while studying at Coventry University in the UK and have been fully immersed in the digital world ever since. In 2009, I founded Visibility, which has grown into one of the leading digital agencies in Slovakia and the Czech Republic, with 100 colleagues across our Bratislava and Prague offices. Today, I serve as founder and chairman of the Visibility 360 Group (https://visibility360.group/en/) – an umbrella brand uniting 8 agencies and 140+ experts across marketing, technology, and consulting.

Peter Šebo, *Founder & CDO*, PS Digital

The implementation of the EU AI Act in Slovakia is progressing in a cautious, institution-first way: the main focus so far is on designating competent authorities, building a supervisory framework, and linking AI regulation to the broader 2030 digital transformation strategy, rather than rushing into a sweeping standalone national AI law. In practice, this means that organisations are currently more concerned with who in the state will supervise what, and how compliance will be enforced, than with highly detailed technical norms. Sector-specific "soft-law" instruments are emerging, for example ethical guidelines for teachers, internal university directives on AI use, and proposals from the ASAI Association, which are already shaping day-to-day expectations around transparency, human oversight and responsible AI use in education, business and legal services, even before full-scale formal supervision is in place.

Biography

Peter Šebo, is a digital marketer and marketing enthusiast. Founder of series of unique digital marketing conferences in Slovakia and the first national internet and mobile apps awards in the country. Organizer of Marketing RULEZZ, the biggest Slovak marketing conference. Regularly lecturing at professional conferences and workshops and several Slovak universities about digital marketing, social media and AI. He also co-hosts the most successful Slovak marketing podcast, Buzzworld, and leads his own digital marketing consultant agency, PS:Digital.

Appendix B: OECD AI Progress Report

This note summarises the OECD country note on **Slovakia's** progress implementing the EU Coordinated Plan on Artificial Intelligence, and how its AI-relevant commitments are embedded within broader national strategies on digitalisation, data, infrastructure and skills.

National Vision & Strategy

- Slovakia does *not* yet have a standalone national AI strategy. Rather, AI policy is embedded within its broader digital and data strategies.
- Key strategic documents include:
  - The **2030 Digital Transformation Strategy** (adopted by MIRRI), which addresses emerging technologies such as AI, big data, 5G, HPC, etc.
  - The **Action Plan for Digital Transformation 2023–2026**, which contains a dedicated pillar for AI/digital society, with measures, targets, KPIs, and budget.
  - The **National Digital Skills Strategy & Action Plan 2023–2026**, aimed at improving digital literacy, boosting ICT specialists, and reducing digital exclusion.

- The **National Digital Decade Strategic Roadmap**, which aligns with the EU's Digital Decade goals and harmonises infrastructure, skills, business digitalisation and public services.

- On governance, Slovakia has established coordinating bodies including:
  - A **Standing Commission on Ethics and Regulation of AI (CERAI)**, which advises on ethical, legal, and social implications.
  - **AISlovakIA**, a non-profit platform uniting academia, business, government, and civil society around AI development.
  - Institutional proposals for a **Government Council for Digitalisation**, a **Task Force for Digital Governance**, and a policy architecture (Digital Policy Map, Digital Policy Forum, Expert Directory) to foster interagency and cross-sector coordination.

Data & Infrastructure

- Slovakia is building enabling conditions for AI through data infrastructure and computing capacity: its digital strategies describe ambitions around cloud, edge computing, and HPC.
- There is strong policy support for pooling data, improving data quality, and making data more accessible in formats useful for AI.
- On compute: while specific HPC projects are mentioned, the strategic documents show awareness of the need for critical compute to support advanced AI workloads.

Research, Innovation & Market

- Research capacity is supported via several initiatives:
  - The **Slovak Center for Artificial Intelligence Research** (AISlovakia) — a platform connecting industry, academia, government, and start-ups.
  - Regional AI hubs in universities, which localise AI research closer to industry and regional needs.
  - **European Digital Innovation Hubs (EDIHs)** are active in Slovakia, including ones focused on healthcare innovation, to help organisations adopt AI and other tech.

Skills & Education

- Education reforms are underway to build a pipeline of AI-ready citizens:
  - Primary and secondary education curriculum is being reformed to embed digital competences, including AI-relevant knowledge.
  - A substantial training component is planned for school staff (digital coordinators, teachers) to help with digital/AI pedagogy.
  - Vocational training and tertiary education are also supported: universities teach AI subjects, and non-formal training is encouraged to upskill the workforce.

Sector Deployment & Public Use

- Public Sector / Governance:
  - Slovakia is preparing an **AI sandbox** ("Experimental Regulatory Environments for AI") to test AI applications under controlled regulatory conditions. Preparatory work (legislative analysis, methodology) is slated for 2025.
  - There is a newly launched **information hub for AI**, co-developed with AISlovakIA, to centralise legislation, initiatives, and resources for both experts and the general public.
- Healthcare:
  - Slovakia has a mature health-data infrastructure: its National eHealth system (including electronic health records) is centralised via the NCZI, enabling data access and possibly secondary use for research/AI.
  - There are efforts to improve data interoperability and make health data available for AI-based research under controlled, secure environments.
- Mobility:
  - AI is being integrated into transport planning. For example, the Slovak Strategic Transport Development Plan references intelligent transport systems (ITS) and real-time data systems for road, rail, and waterways.

- Environment / Climate:
    - AI is also being used in environmental applications: the Slovak Republic mentions AI in its strategies for climate, sustainability, and digital transformation.

Trust, Governance & Ethics

- Ethical governance is being institutionalised: via CERAI (expert ethics commission) which helps guide regulation, risk considerations, and ethical adoption of AI.
- Coordination gaps remain: while many strategies exist, they are still being harmonised (e.g., aligning AI policy embedded in digital strategies under a more coherent framework).
- Proposals for a Government Council for Digitalisation and a Task Force for Digital Governance suggest Slovakia recognises the need for stronger cross-ministry oversight.

3. Strengths

- **Integrated, systemic approach**: Rather than a siloed AI strategy, Slovakia embeds AI across its digital, data, infrastructure, and education strategies — this ensures coherence and alignment with broader national priorities.
- **Governance institutions**: The establishment of bodies such as CERAI and AISlovakIA enhances ethical oversight, multi-stakeholder dialogue, and long-term policy coordination.
- **Data and compute readiness**: The strategic inclusion of cloud, edge, and HPC in its digital policies signals serious commitment to AI-enabling infrastructure.
- **Strong research and innovation base**: Via AISlovakia, regional hubs, and EDIHs, Slovakia is well positioned to translate academic capacity into innovation and economic deployment.
- **Education pipeline for digital and AI literacies**: Reforming curricula, empowering teachers, and increasing vocational/reskilling opportunities builds long-term human capital.
- **Pilot-friendly public sector**: The planned AI sandbox suggests Slovakia is ready to test AI use-cases in public services under regulatory supervision.

4. Constraints and Risks

- **Absence of a dedicated AI strategy**: Without a standalone AI strategy, there is a risk of fragmentation, where AI goals are diluted across multiple documents, making accountability weaker.
- **Coordination challenges**: Multiple strategies (digital, skills, infrastructure) need harmonisation; until that is achieved, siloing across ministries could hamper coherent implementation.
- **Resource and capacity limitations**: While ambitions for HPC, cloud, edge are present, implementing them (and ensuring SMEs/public sector can use them) may be constrained by funding, technical capacity, or demand.
- **Trust and public engagement risk**: Ethical oversight exists, but deploying AI in sensitive domains (e.g., health) requires sustained public engagement, transparency, and robust risk governance.
- **Skills bottleneck**: Even with a skills strategy, demand for AI specialists may outstrip supply, especially for advanced roles (data scientists, AI researchers).
- **Scaling adoption**: Pilot projects (e.g., via sandboxes) need to be translated into scalable, sustainable operations; otherwise, the impact may remain limited.
- **Under-utilisation of data assets**: Health data infrastructure is promising, but managing access, interoperability and secondary use in practice may be challenging.

5. Practical Recommendations (Next Steps)

1. **Develop a dedicated national AI strategy**
    - Convene a working group (with government, academia, industry, civil society) to craft a standalone AI strategy, building on existing digital and data strategies.
    - Define clear KPIs, timelines, budget, and assigned lead agencies to ensure accountability.

2. **Strengthen cross-ministerial coordination**
    - Fully operationalise the proposed Government Council for Digitalisation and the Task Force for Digital Governance.
    - Create a unified "Digital Policy Forum" to align AI, data, skills, and infrastructure policies.
3. **Expand compute capacity access**
    - Ensure HPC and cloud resources are not just for large institutions, but accessible to SMEs, start-ups and public agencies.
    - Set up fellowship or accelerator programmes that allow smaller organisations to leverage compute to build AI applications.
4. **Operationalise the AI sandbox**
    - Move swiftly on the 2025 preparatory phase: finalize legal, technical, and operational frameworks.
    - Use the sandbox to run real pilot projects in public administration, healthcare, mobility, etc. Collect lessons and scale successful pilots.
5. **Embed trust and ethics in deployment**
    - Strengthen CERAI's mandate to provide not only advisory input but also monitor deployed AI systems.
    - Launch a public-facing AI observatory or report that publishes risk assessments, ethical reviews, and sectoral deployments to build transparency.
6. **Invest in skills at all levels**
    - Roll out teacher training and resourcing (methodologies, textbooks) to embed AI literacy in primary / secondary education.
    - Expand vocational and micro-credential programmes in AI, data science, and digital infrastructure.
    - Partner with industry to create upskilling programmes, apprenticeships, and return-to-research fellowships in AI / HPC.

7. **Scale sector pilots into real deployments**

    - Use EDIHs and research hubs to run pilot AI projects in healthcare, transport, environment, and public services.
    - Support scaling via matched funding, public procurement incentives, and publicly-led “first buyer” schemes.

6. Overall Assessment

Slovakia has made **notable progress** in establishing the foundations of an AI-enabling ecosystem, even without a standalone AI strategy. Its approach is **integrated and holistic**, embedding AI within its broader digital transformation, skills development, data infrastructure, and governance frameworks. This gives it flexibility and coherence, but the absence of a dedicated AI strategy also poses risks of fragmentation and diluted accountability.

The strengths lie in its coordinated governance (CERAI, AISlovakIA), a clear focus on skills from school to lifelong learning, and ambition around compute infrastructure. However, to translate strategy into impact, Slovakia must focus on execution: ensuring cross-sector coordination, making compute capacity accessible, scaling pilot programmes, and embedding trust mechanisms in real-world deployments.

If Slovakia can accelerate the establishment of its AI sandbox, operationalise governance structures, and promote inclusive access to infrastructure and skills, it is well-positioned to leverage AI for economic innovation and public-good applications.

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This section compiles all primary and secondary sources cited—legal texts, policy papers, academic studies, and official reports. It provides a transparent basis for the report's analyses and recommendations, enabling verification, further research, and continued dialogue on AI policy and governance in Slovakia.

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